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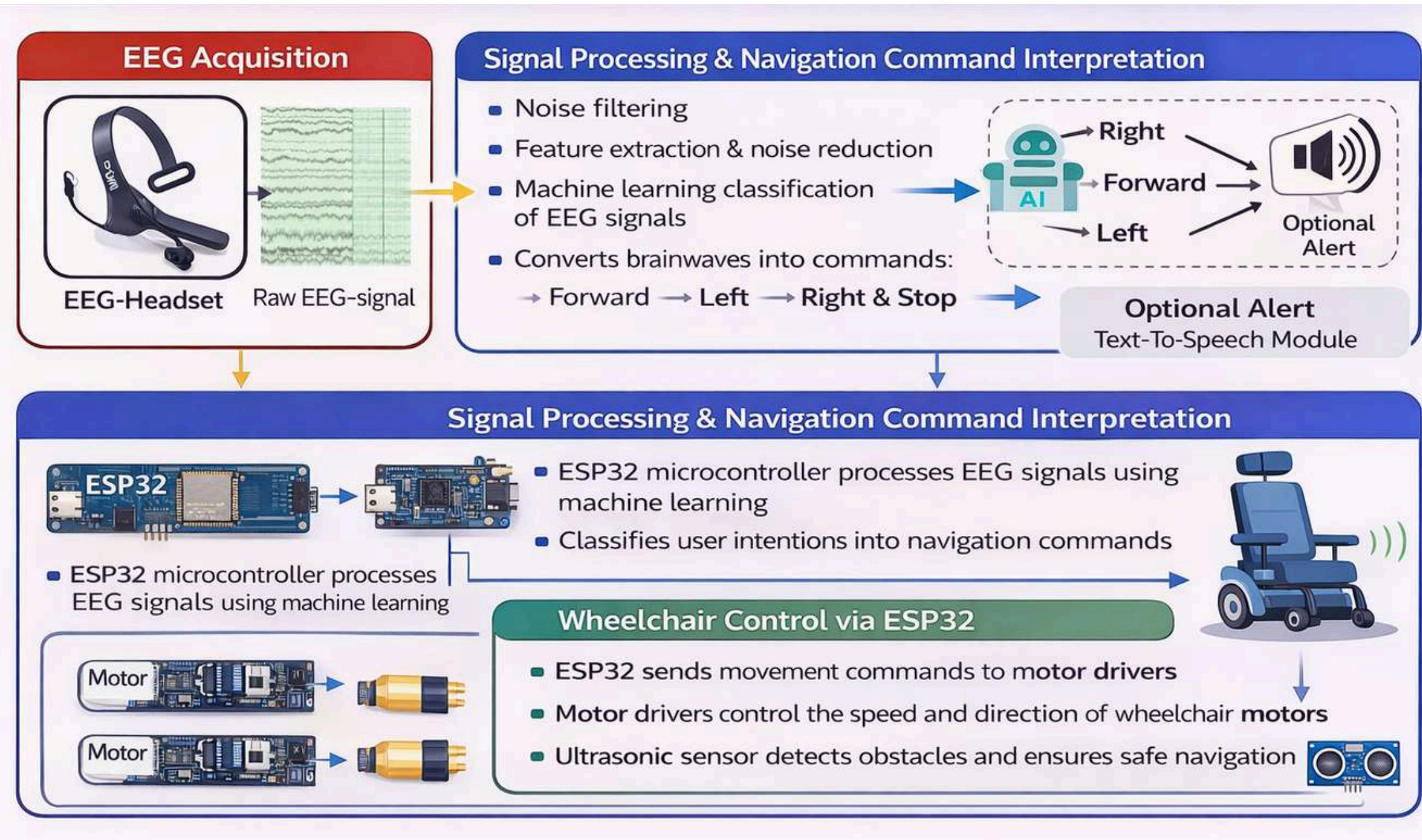


Brain Controlled Verbal Communication Enabled Wheelchair

Proposed Solution:

How Our Project Helps Disabled People

- Many people with severe disabilities cannot use their hands or legs to control a wheelchair, making independent movement difficult.
- Our project develops a **brain-controlled smart wheelchair** that allows users to control movement through **eye blinks detected by an EEG headset**.
- The EEG headset captures **brain and eye-blink signals** and sends them to the processing system.
- Different **eye blink patterns** are interpreted as **commands** such as **Forward, Left, Right, or Stop**.
- These commands are sent to the **ESP32 controller**, which moves the wheelchair in the required direction.
- An **ultrasonic sensor** detects obstacles and **stops the wheelchair** automatically to ensure user safety.
- The system also **displays live brain signals** on a **monitoring portal**, allowing doctors or caregivers to observe system activity in real time.





TECHNICAL APPROACH

Hardware Components



EEG Acquisition

Device: NeuroSky MindWave Headset

Purpose: Capture brain signals for user intention detection



Ultrasonic Sensor

Device: HC-SR04 Ultrasonic Sensor

Purpose: Detect obstacles in front of the wheelchair
• Prevent collisions • Automatic stop for safety



Amplification & Filtering

IC: AD620 / INA123P Instrumentation Amplifier

Purpose: Amplify and filter weak EEG signals to remove noise



Microcontroller

Board: ESP32 DevKit V1

Purpose: Receives EEG commands via Bluetooth, Processes navigation, Controls motors



Motors

Device: DK36 – 100K DC Motor

Purpose: Control direction and speed



Wheelchair

Type: Medical Grade Wheelchair

Purpose: Base mobility platform



Software & Programming Tools



Signal Processing

• **Python** + pyminwave2

Used for:

- ✓ EEG signal acquisition
- ✓ Brainwave preprocessing



Machine Learning

• **PyTorch** / scikit-learn

Used for:

- ✓ Signal classification
- ✓ Navigation command prediction



Embedded Control

• **ESP32** Firmware

Functions:

- ✓ Motor control
- ✓ Command execution
- ✓ System communication



Speech Communication

• **ESP32** + Google TTS (online)

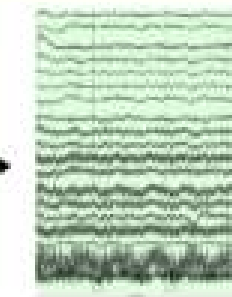
Used for:

- ✓ Converting text to speech
- ✓ Assistive voice feedback

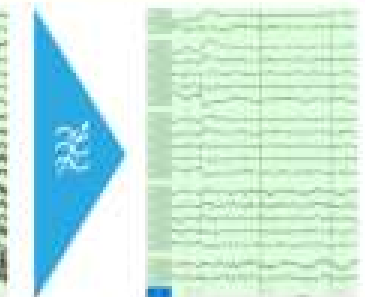
EEG Acquisition



EEG-Headset

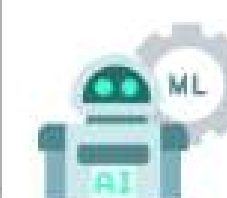


Raw EEG-signal



Processed EEG-signal

Signal Processing & Navigation Command Prediction



Right

Forward

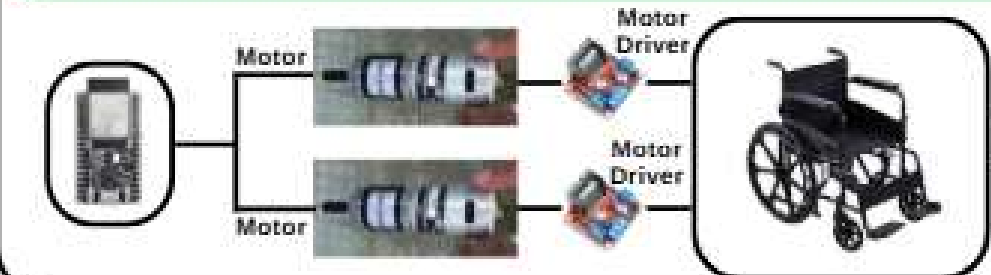
Left

Mapping the navigation commands to motor signals



Text-To-Speech Module

Wheelchair Control via ESP32



Wheel-Chair

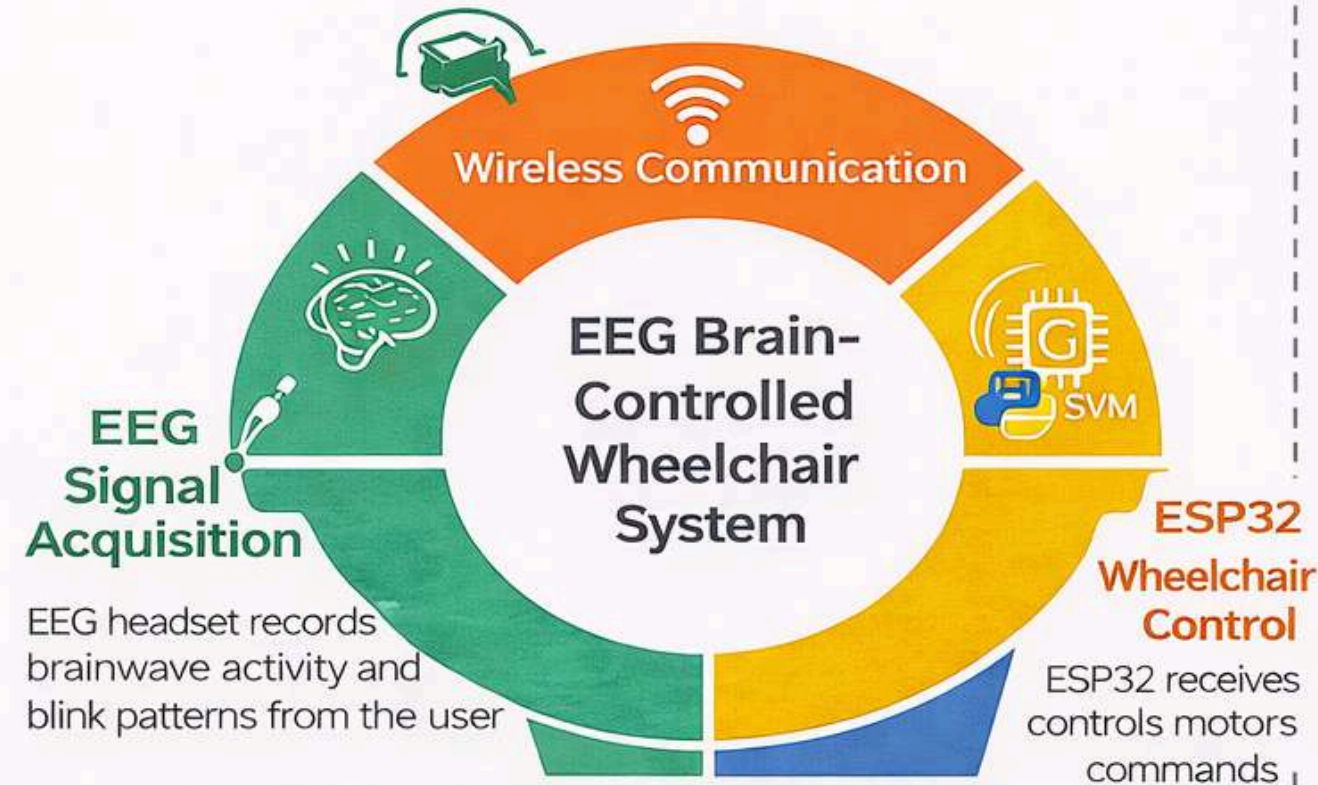


FEASIBILITY AND VIABILITY

Technical Feasibility

EEG Signal Acquisition

EEG headset records brain wave activity and blink patterns from the user



Intelligent Brain-Driven Wheelchair Navigation

Challenges

- EEG signal noise
- Blink pattern variability
- Real-time processing delay



Advantages

- Hands-free wheelchair control
- Improved independence for disabled users

Intelligent Brain-Driven Wheelchair Navigation

Challenges and Risks

Signal Noise

Weak EEG signals can affect command detection

ML Model Reliability

Limited training data may reduce prediction accuracy

User Adaptation

Different users require calibration for blink commands



Strategies to Overcome Challenges

Signal Noise

Weak EEG signals can affect command detection

ML Model Reliability

Limited training data may reduce prediction accuracy

System Latency

Processing delays can slow wheelchair response

User Adaptation

Different users require calibration for blink commands

Strategies to Overcome Challenges

Signal Processing

Advanced filtering and artifact removal

Model Improvement

Continuous training of ML classification model

Real-Time Optimization

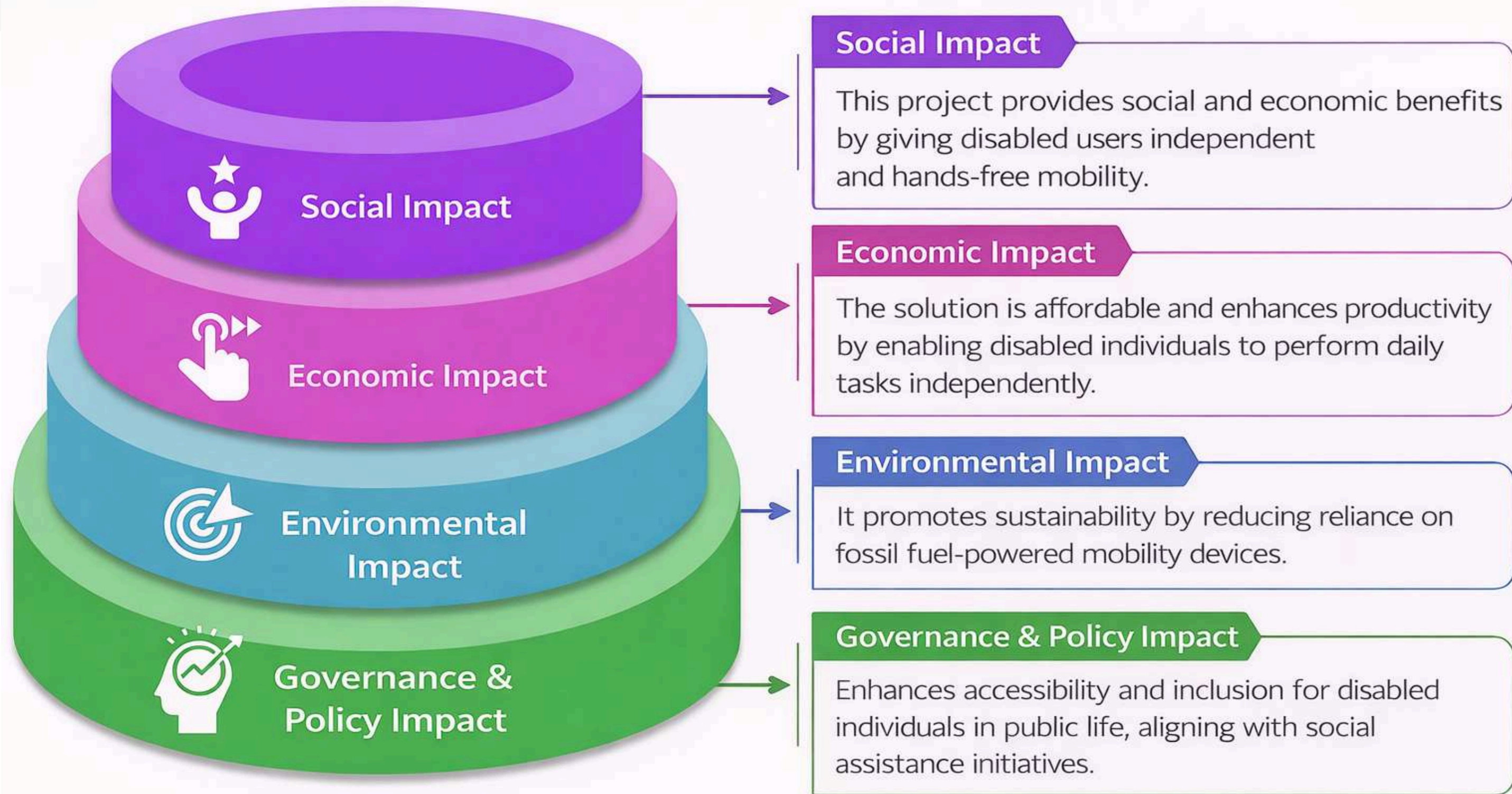
Efficient processing pipeline for faster response

Hardware Stability

Reliable ESP32 communication and motor control



IMPACT AND BENEFITS





RESEARCH & REFERENCES

Our References:

- [1] A. Gupta, S. Pathan, P. Sarmal, and O. Rastogi, “Brain Computer Interface (BCI)-Inspired Arduino Based Wheel Chair Controller,” in Proc. 2023 Int. Conf. Adv. Comput. Commun. Technol. (ICACCTech), 2023.
- [2] E. V. K. Rao, N. Y. R. Reddy, B. V. S. S. Greeshma, and Y. S. S. V. Reddy, “EEG Based Smart Wheelchair For Disabled Persons Using Non-Invasive BCI,” in Proc. 2022 Int. Conf. Comput. Intell. Sustain. Eng. Solutions (CISES), 2022.
- [3] K. Alsammarrarie and T. Inan, “Car Control by using brain waves and Arduino based Mind wave Mobile,” in Proc. 2022 Int. Congr. Human-Computer Interaction, Optimization and Robotic Applications (HORA), 2022.
- [4] L. Rahman, J. Adan, Q. M. Billah, M. K. Islam, A. H. M. M. Kamal, S. Hossain, C. Shahnaz, and A. Subhana, “A low cost Human Computer Interface for Disabled People based on Eye Blink detection using Brain Signal,” in Proc. TENCON 2019 – 2019 IEEE Region 10 Conf. (TENCON), 2019.
- [5] P. Lahane, S. P. Adavardkar, S. V. Tendulkar, B. V. Shah, and S. Singhal, “Innovative Approach to Control Wheelchair for Disabled People Using BCI,” in Proc. 2018 3rd Int. Conf. Conver. Technol. (I2CT), 2018.

SR NO.	TITLE OF PAPER	AUTHOR NAME	WHERE AND WHEN PUBLISHED	WHAT IS IMPLEMENTED	WHAT CAN BE DONE
1	Smart and Automatic Wheelchair using Arduino Uno	Ketan.P.Margale Dhananjay.S.Matkar Ajit.H.Sarak	International Journal of Scientific Research in Engineering and Management (IJSREM) Volume: 09 Issue: 04 April - 2025	- A smart, gesture-controlled wheelchair using Arduino Uno/Mega, ADXL335 accelerometer, RF modules, ultrasonic sensors, and relay motor drivers. - Hand tilts (X/Y axis) interpreted into direction commands (FWD/BWD/LEFT/RIGHT) with ~90-95% recognition accuracy.	- Integrate machine learning for adaptive, personalized gesture recognition. - Add multimodal control (head/eye tracking, voice, BCI). - Improve real-time performance, power efficiency, and expand user testing for reliability. - Explore GPS tracking and emergency alerts.
2	An IoT-Based Smart Wheelchair with EEG Control and Vital Sign Monitoring	Rowida Meligy, Anton Royanto Ahmad and Samir Mekid	Presented at the 11th International Electronic Conference on Sensors and Applications (ECSA-11), 26-28 November 2024	- IoT-enabled smart wheelchair integrating EEG-based BCI, joystick, vital sign monitoring (ECG, SpO2, heart rate), tri-wheel stair-climbing mechanism, and cloud connectivity. - EEG signal classification using SVM on Jetson Nano. - Emergency alerts via SIM808 module.	- Conduct extensive user trials for usability across different impairments. - Improve ML models for PVC and anomaly detection. - Miniaturize and optimize design for cost, portability, and mass deployment. - Integrate more sensors and connect with smart city infrastructure.
3	A Low Cost Human Computer Interface for Disabled People Based on Eye Blink Detection Using Brain Signal	Lamiya Rahman*, Jannatul Adan*, Quazi Mutasim Billah, Md Kamrul Islam, A.H.M Mostafa Kamal, Sazzad Hossain, Dr Celia Shahnaz and Arik Subhana	2019 IEEE Region 10 Conference (TENCON 2019)	A portable EEG-based blink detection system using low-cost sensors (NeuroSky, Muse) and threshold-based signal analysis enables GUI-based control of devices. With ~98-99% accuracy tested on 25 subjects, the Arduino-powered system integrates EEG sensors, amplifiers, and display hardware mounted on a wheelchair.	Extend testing to real disabled users, enhance accuracy with adaptive thresholding, and add more control options with broader device support. Integrate wireless EEG modules, improve portability, reduce false positives, and miniaturize hardware for efficient daily use.
4	Inovative Approach to Control Wheelchair For Disabled People Using BCI	Prashant Lahane, Swarada Pushkar Adavardkar, Sarvesh Vilas Tendulkar, Bhoomi Viralkumar Shah,Shubham Singhal	2018 3rd International Conference for Convergence in Technology (I2CT) The Gateway Hotel, XION Complex, Wakad Road, Pune, India. Apr 06-08, 2018	A cost-effective, user-friendly BCI system using NeuroSky™ MindWave detects eye blinks, processed via a Python app to wirelessly control a wheelchair through Arduino and L298N. It enables real-time, hands-free navigation with ~90% accuracy and minimal cognitive effort.	Enhance the system using multi-channel EEG, refined blink detection, and additional brain signals for nuanced control and safety features like obstacle detection. Focus on hardware miniaturization, support for complex commands, and long-term trials across diverse users for real-world validation.